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LED frame fixture

Abstract

A light fixture includes a frame that includes a plurality of frame pieces configured for assembly around a ceiling tile to form the light fixture. Each of the plurality of frame pieces include an L-shaped diffuser and a circuit board that includes a plurality of light sources to emit illumination. The L-shaped diffuser includes a diffuser body that is asymmetric in transparency to direct more illumination from the light sources to inside the ceiling tile than outside the ceiling tile. The diffuser body includes an outside wall that is substantially opaque, an inside wall that is at least partially substantially transparent, and a bottom wall that is substantially transparent. The light fixture can further include a connector to connect adjoining frame pieces together, which includes a substantially opaque portion below the circuit board and a substantially transparent portion above to allow illumination to pass between the adjoining frame pieces.

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Background/Summary

TECHNICAL FIELD
(1) The present subject matter is drawn to technologies that improve the aesthetics and light output of a luminaire, such as a light fixture. In particular, the disclosed technologies focus on improving light output from the light fixture, and reducing or even eliminating shadows that can occur at the interface between two connected portions of the light fixture.
BACKGROUND
(2) Electrically powered artificial lighting for general illumination has become ubiquitous in modern society. Electrical lighting equipment is commonly deployed, for example, in homes, buildings of commercial and other enterprise establishments, as well as in various outdoor settings.
(3) In conventional luminaires, such as light fixtures, the luminance output can be turned ON/OFF and often can be adjusted up or dimmed down. In some light fixtures, e.g., using multiple colors of light emitting diode (LED) type sources, the user may be able to adjust a combined color output of the resulting illumination. The changes in intensity or color characteristic of the illumination may be responsive to manual user inputs or responsive to various sensed conditions in or about the illuminated space.
(4) An LED frame fixture is a type of light fixture that typically includes several frame pieces (e.g., linear channels) that can be assembled around 1×4, 2×2, or 2×4 foot ceiling tiles, for example. Typically, the LED frame fixture includes four frame pieces with LEDs and can be wrapped around square or rectangular shaped ceiling tiles in order to put LED lighting on a drop ceiling. The LED frame fixture is suitable for retrofitting a room with light fixtures or new construction installations.
(5) LED frame fixtures include light sources that emit illumination and often include a diffuser to spread out or scatter light output from the light sources. Conventional LED frame fixtures in the market have a U-shaped diffuser that resembles the letter U in cross-sectional profile. The U-shaped diffuser causes an appearance of high glare, especially at a high angle, to an observer of the illumination emitted from the LED frame fixture. Also, conventional LED frame fixtures have shadows or breaks in illumination at the corner where the two frame pieces connect to each other, which does not look appealing to the observer of the illumination from below.
(6) Accordingly, light fixtures, lighting devices, and a lighting system are needed to overcome these and other limitations in the art.
SUMMARY
(7) In a first example, a light fixture **101** includes a frame **105** that includes a plurality of frame pieces **107**A-N configured for assembly around a ceiling tile **109** to form the light fixture **101**. Each of the plurality of frame pieces **107**A-N include an L-shaped diffuser **111** and a circuit board **113** that includes a plurality of light sources **115**A-N to emit illumination **117**. The L-shaped diffuser **111** includes a diffuser body **119** that is asymmetric in transparency to direct more illumination **117** from the light sources **115**A-N to inside **121** the ceiling tile **109** than outside **123** the ceiling tile **109**. The diffuser body **119** includes an outside wall **125** that is substantially opaque, an inside wall **127** that is at least partially substantially transparent, and a bottom wall **129** that is substantially transparent.

(8) In a second example, a light fixture **101** includes a frame **105** that includes a plurality of frame pieces **107**A-N configured for assembly around a ceiling tile **109** to form the light fixture **101**. Each of the plurality of frame pieces **107**A-N include an L-shaped diffuser **111** and a circuit board **113** that includes a plurality of light sources **115**A-N to emit illumination **117**. The L-shaped diffuser **111** includes a diffuser body **119** that is asymmetric in transparency to direct more illumination **117** from the light sources **115**A-N to inside **121** the ceiling tile **109** than outside **123** the ceiling tile **109**. The light fixture **101** further includes a connector **135** to connect adjoining frame pieces **107**A-B of the plurality of frame pieces **107**A-N together. The connector **135** includes a substantially opaque portion **137** below the circuit board **113** and a substantially transparent portion **139** above the circuit board **113** to allow illumination **117** to pass between the adjoining frame pieces **107**A-B.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawing figures depict one or more implementations in accord with the present teachings, by way of example only, not by way of limitation. In the figures, like reference numerals refer to the same or similar elements.

(2) FIG. **1**A depicts a lighting system that includes a plurality of light fixtures located in a space, such as a room or a building.

(3) FIG. **1**B is an isometric view of a lighting system depicting a light fixture assembled around a ceiling tile of a ceiling tile grid.

(4) FIG. **1**C is another isometric view of the lighting system depicting a lighting device that includes the light fixture.

(5) FIG. **2**A is a bottom view of the light fixture, with an encircled detail area to show context for the cutaway view of FIG. **2**B of adjoining frame pieces and an L-shaped diffuser.

(6) FIG. **2**B is a cutaway view of the encircled detail area of FIG. **2**A and shows details of adjoining frame pieces, a circuit board, light sources, and the L-shaped diffuser.

(7) FIG. **2**C is a cutaway view of the adjoining frame piece of FIG. **2**B showing details of the inside wall of the diffuser body of the L-shaped diffuser.

(8) FIG. **3**A is an isometric view of the adjoining frame piece, with an encircled detail area to show context for the cutaway views of FIGS. **3**B-C of a connector between adjoining frame pieces.

(9) FIGS. **3**B-C are cutaway views of the encircled detail area of FIG. **3**A and show details of the connector.

(10) FIG. **4**A is an isometric view of the adjoining frame piece, with an encircled detail area to show context for the cutaway view of FIG. **4**B of a male connector between adjoining frame pieces **107**A-B.

(11) FIG. **4**B is a zoomed in view of the encircled detail area of FIG. **4**A and shows details of the male connector.

(12) FIG. **5**A is a bottom view of the light fixture, with an encircled detail area to show context for the views of FIG. **5**B-D of the adjoining frame pieces, circuit board, and the connector.

(13) FIG. **5**B is a cutaway view of the encircled detail area of FIG. **5**A and shows details of the adjoining frame pieces, circuit board, light sources, and the connector.

(14) FIG. **5**C is a zoomed in view of the connector that includes a substantially transparent portion and a substantially opaque portion.

(15) FIG. **5**D is an isometric view of the assembled light fixture with the connector connecting the adjoining frame pieces together.

(16) FIG. **6** is a cross-sectional view of the adjoining frame piece depicting details of the circuit board, illumination light source, and a chamfer.

PARTS LISTING

(17) **100** Lighting System **101** Light Fixture **102**A-N Lighting Control Devices **103**A-N Sensors **105** Frame **107**A-N Frame Pieces **107**A-B Adjoining Frame Pieces **109**, **109**A-N Ceiling Tiles **111** L-Shaped Diffuser **113** Circuit Board **115**A-N Light Sources **115**A-B Extra Light Sources **117** Illumination **119** Diffuser Body **121** Inside **123** Outside **125** Outside Wall **127** Inside Wall **129** Bottom Wall **131** Substantially Transparent Portion **133** Substantially Opaque Portion **135**, **135**A-N Connectors **136**, **136**A-N Male Connectors **137** Substantially Opaque Portion **139** Substantially Transparent Portion **141** Opening **143** Light Guide **145** Corner **151** Chamfer **153** Angled Surface **155** Highly Reflective Surface **157** Interior Cavity **160** Lighting Device **161** Driver Circuit **163** Cable **167** Ceiling Tile Grid **170** Space **172** Observer **200** Detail Area **300** Detail Area **400** Detail Area **500** Detail Area

DETAILED DESCRIPTION

(18) In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. However, it should be apparent to those skilled in the art that the present teachings may be practiced without such details. In other instances, well known methods, procedures, components, and/or circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present teachings.

(19) Reference now is made in detail to the examples illustrated in the accompanying drawings and discussed below.

(20) FIG. **1**A depicts a lighting system **100** that includes a plurality of light fixtures **101**A-N located in a space **170**, such as a room or a building. In the example, light fixtures **101**A-N illuminate the space **170** of a premises to a level useful for an observer **172**, such as a human, in or passing through a location in the space **170**, such as a warehouse, room, or a corridor in a building; or of an outdoor space such as a street, sidewalk, parking lot or performance venue. Lighting control devices **102**A-N can be wall switches or touch screen devices to turn on/off or dim light fixtures **101**A-N. Occupancy, daylight, or audio sensors **103**A-N can enable controls for on/off, occupancy, and dimming of the light fixtures **101**A-N.

(21) FIG. **1**B is an isometric view of a lighting system **100** depicting a light fixture **101** assembled around a ceiling tile **109** of a ceiling tile grid **167**. The light fixture **101** is illustrated from the perspective of the observer **172** looking up at the ceiling from the floor of the space **170**. A light fixture **101** for general illumination **117** of the space **170** like that constructed herein with an L-shaped diffuser **111** has illumination **117** with an appearance of a soft glow on the ceiling tile **109**. The soft glow of the illumination **117** can be provided by the uplight enabled by the L-shaped diffuser **111**.

(22) The light fixture **101** includes a frame **105** that includes a plurality of frame pieces **107**A-N configured for assembly around a ceiling tile **109** to form the light fixture **101**. Each of the plurality of frame pieces **107**A-N include an L-shaped diffuser **111** and a plurality of light sources **115**A-N(see FIG. **2**B) to emit illumination **117**. After installation of the light fixture **101**, the light sources **115**A-N are oriented at the bottom of the light fixture **101** and down towards the floor of the space **170** to emit output light rays. For example, the L-shaped diffuser **111** helps to homogenize and spread output light rays by diverging incoming light rays emitted by the light sources **115**A-N, for example, to increase a beam angle of the illumination **117**. The L-shaped diffuser **111** is asymmetric in transparency to direct more illumination from the light sources **115**A-N to inside **121** the ceiling tile **109** than outside **123** the ceiling tile **109**.

(23) In the example (see FIG. **2**B), a diffuser body **119** of the L-shaped diffuser **111** blocks the illumination **117** from an outside wall **125** to reduce excessive glare and light emitted outside of the ceiling tile **109** by the assembled light fixture **101**. The diffuser body **119** of the L-shaped diffuser **111** allows the illumination **117** to be emitted from the bottom wall **129** and shoot down towards the floor to light up the floor of the space **170**. The diffuser body **119** also allows the illumination

117 to be emitted from the inside wall **127** to light up inside **121** of the ceiling tile **109** and appear more decorative. In contrast, conventional LED frame fixtures provide an appearance of a cave on the ceiling tile **109**.

(24) FIG. **1**C is another isometric view of the lighting system **100** depicting a lighting device **160** that includes the light fixture **101**. The lighting device **160** is illustrated from the perspective of the observer **172** looking down at the floor from the ceiling of the space **170**. The lighting device **160** further includes a driver circuit **161** to control the light sources **115**A-N(see FIG. **2**B) of the light fixture **101**. The lighting device **160** can further include a cable **163** configured to connect via electrical connection to the circuit board **113** (see FIG. **2**B) of the light fixture **101** and the driver circuit **161** to control the light sources **115**A-N(see FIG. **2**B). As shown, the driver circuit **161**, such a driver box, is configured to mount on a ceiling tile grid **167** and is mounted separate from the light fixture **101** assembled around the ceiling tile **109**. The driver box can include clips and may be held in place on the ceiling tile grid **167** by snapping the clips onto the ceiling tile grid **167**. For example, the cable **163** can be an FD4 cable that comes out of the light fixture **101** at a corner **145** (see FIG. **5**B) to connect to the driver circuit **161** (e.g., driver box). In some examples, the light fixture **101** may include a ballast instead of the driver circuit **161** depending on the type of illumination light sources **115**A-N(e.g., for a fluorescent or incandescent light bulb).

(25) Ceiling tiles **109**A-N, also known as ceiling panels, can formed of lightweight construction materials to cover a ceiling. The ceiling tiles **109**A-N can be placed in a ceiling tile grid **167** (T-grid) to improve the aesthetics or acoustics of room and provide some thermal insulation. The ceiling tile grid **167** is a frame for suspended ceilings, also known as drop ceilings, that utilize ceiling tiles **109**A-N. The ceiling tile grid **167** hangs from the main structural ceiling or beams and can be fastened to the walls of a room to support the ceiling tiles **109**A-N to finish the space **170**. For example, the ceiling tile grid **167** can include steel or aluminum bars, such as a main beam, cross tees, and wall molding, to support the ceiling tiles **109**A-N. The ceiling tiles **109**A-N fit in spaces between the ceiling tile grid **167** to complete the ceiling and may be held in place by panel clips to the ceiling tile grid **167**.

(26) FIG. **2**A is a bottom view of the light fixture **101**, with an encircled detail area **200** to show context for the cutaway view of FIG. **2**B of adjoining frame pieces **107**A-B and an L-shaped diffuser **111**. As shown, the light fixture **101** includes a frame **105** that includes a plurality of frame pieces **107**A-N configured for assembly around a ceiling tile **109** to form the light fixture **101**.

(27) L-shaped diffuser **111** can control a light distribution of the illumination **117** while trying to maximize optical efficiency of the light output. This improves uniformity of illumination light output intensity of the illumination **117**, as may be perceived from a working distance from the light fixture **101**, such as by the observer **172** walking on the floor of the space **170** or sitting at a desktop. The L-shaped diffuser **111** can be a film formed of a polycarbonate (PC) substrate, where the substrate typically has an index of refraction of 1.58. If the L-shaped diffuser **111** is formed of acrylic (e.g. PMMA), the substrate typically has an index of refraction of 1.49. If the L-shaped diffuser **111** is formed of low-density polyethylene (LDPE or just PE), the substrate typically has an index of refraction of 1.51.

(28) FIG. **2**B is a cutaway view of the encircled detail area **200** of FIG. **2**A and shows details of adjoining frame pieces **107**A-B, a circuit board **113**, light sources **115**A-N, and the L-shaped diffuser **111**. As depicted, each of the plurality of frame pieces **107**A-N include an L-shaped diffuser **111** and a circuit board **113** that includes a plurality of light sources **115**A-N to emit illumination **117**. The L-shaped diffuser **111** includes a diffuser body **119** that is asymmetric in transparency to direct more illumination **117** from the light sources **115**A-N to inside **121** the ceiling tile **109** than outside **123** the ceiling tile **109**. Compared to conventional symmetric U-shaped diffusers, the L-shaped diffuser **111** gives more light to the inside **121** of the ceiling tile **109** compared to the outside **123** of the ceiling tile **109**. For example, the diffuser body **119** can illuminate the edge and the middle of the ceiling tile **109**, but not outside **123** the ceiling tile **109**.

(29) The diffuser body **119** includes an outside wall **125** that is substantially opaque, an inside wall **127** that is at least partially substantially transparent, and a bottom wall **129** that is substantially transparent. For example, the inside wall **127** is divided into an upper portion and a lower portion. The upper portion is substantially opaque at the top part of the inside wall **127** (substantially opaque portion **133**) to get light out of the lower portion. The lower portion is substantially transparent at the bottom part of the inside wall **127** (substantially transparent portion **131**) so it allows up light to the ceiling tile **109** is an uplight configuration. Although the entire diffuser has three walls that form a U-shape, the L-shaped diffuser **111** is said to be L-shaped because the outside wall **125** is an extrusion that is substantially opaque.
(30) Light sources **115**A-N include electrical-to-optical transducers, such as various light emitters for illumination **117**. The emitted light may be in the visible spectrum or in other wavelength ranges. Suitable light generation sources for light sources **115**A-N include various conventional lamps, such as incandescent, fluorescent or halide lamps; one or more light emitting diodes (LEDs) of various types, such as planar LEDs, organic LEDs (OLEDs), micro LEDs, micro OLEDs, LEDs on gallium nitride (GaN) substrates, micro nanowire or nanorod LEDs, photo pumped quantum dot (QD) LEDs, micro plasmonic LED, micro resonant-cavity (RC) LEDs, and micro photonic crystal LEDs; as well as other sources such as micro super luminescent Diodes (SLD) and micro laser diodes. Of course, these light generation technologies are given by way of non-limiting examples, and other light generation technologies may be used. For example, it should be understood that non-micro versions of the foregoing light generation sources can be used.
(31) FIG. **2**C is a cutaway view of the adjoining frame piece **107**A of FIG. **2**B showing details of the inside wall **127** of the diffuser body **119** of the L-shaped diffuser **111**. The inside wall **127** can be at least partially substantially transparent, such as split into portions so a bottom portion oriented towards the floor of the space **170** is a substantially transparent and a top portion is substantially opaque. Hence, as shown, the inside wall **127** includes a substantially transparent portion **131** (e.g., highly light transmissive) configured to illuminate inside the ceiling tile **109** and a substantially opaque portion **133** to direct illumination **117** to the bottom wall **129**. The outside wall **125** is substantially opaque to block illumination **117** from appearing outside **123** of the ceiling tile **109** and thereby reduce glare. The diffuser body **119** can be co-extruded as a single extrusion of optical material with a white pigment for substantially opaque portions. For example, the optical material is for substantially transparent portions of the diffuser body **119** and can be optically transparent or optically clear. The optical material can be formed of polycarbonate or acrylic. Although a white opaque material is used for substantially opaque portions to block and reflect light from light sources **115**A-N, a different color besides white can be used as desired for a particular layout of the L-shaped diffuser **111**.
(32) FIG. **3**A is an isometric view of the adjoining frame piece **107**A, with an encircled detail area **300** to show context for the cutaway views of FIGS. **3**B-C of a connector **135** between adjoining frame pieces **107**A-B. As shown, the light fixture **101** includes a connector **135** to connect adjoining frame pieces **107**A-B of the plurality of frame pieces **107**A-N together. Specifically, a plurality of connectors **135**A-N can connect the frame pieces **107**A-N together for assembly around the ceiling tile **109**.
(33) To eliminate shadows at the corners (e.g., connection points or edges) between frame pieces **107**A-N, the connectors **135**A-N can be formed of a substantially transparent portion **139** to allow light to pass through and have an opening **141** (e.g., gap or aperture) to guide light. Extra light sources **115**A-B can also be added near the corners to further eliminate shadows. Such a solution minimizes the transition of the frame pieces **107**A-N so the frame **105** looks like one integral piece and the illumination **117** looks more continuous across all of the frame pieces **107**A-N.
(34) FIGS. **3**B-C are cutaway views of the encircled detail area **300** of FIG. **3**A and show details of the connector **135**. As shown, the connector **135** includes a substantially opaque portion **137** below the circuit board **113** and a substantially transparent portion **139** above the circuit board **113** to

allow illumination **117** to pass between the adjoining frame pieces **107**A-B. The substantially transparent portion **139** includes an opening **141** to create a light guide **143** between the adjoining frame pieces **107**A-B. The light guide **143** of the substantially transparent portion **139** is configured to reduce a shadow in a corner **145** (see FIG. **5**B), such as a connection point or edge, where the adjoining frame pieces **107**A-B are connected together. For uniformity of the illumination **117**, the transparency of the substantially transparent portion **139** of the connector **135** can be matched to the transparency of the substantially transparent portions of the L-shaped diffuser **111**, including the bottom wall **129** and the substantially transparent portion **131** of the inside wall **127**.
(35) FIG. **4**A is an isometric view of the adjoining frame piece **107**B, with an encircled detail area **400** to show context for the cutaway view of FIG. **4**B of a male connector **136** between adjoining frame pieces **107**A-B. FIG. **4**B is a zoomed in view of the encircled detail area **400** of FIG. **4**A and shows details of the male connector **136**.
(36) Light fixture **101** can be assembled with four separate frame pieces **107**A-D that an end customer may take out of a package to make the light fixture **101** a square or rectangular shape by attaching connectors **135**A-N, **136**A-N. Connectors **135**A-N, **136**A-N can be a female connector **135** like that depicted in FIGS. **3**A-C or a male connector **136** like that shown in FIGS. **4**A-B. Some of the frame pieces **107**A-N include female connectors **135**A-N and some include male connectors **136**A-N to enable the frame pieces **107**A-N to be joined together. A male connector **136** plugs into the female connector **135** to align with the substantially opaque portion **137** of the female connector **135**. The male connector **136** is for mechanical and electrical connection and may not include an optical portion, such as a substantially transparent portion. There can be an air gap between the opening **141** of the substantially transparent portion **139** of the female connector **135** of the adjoining frame piece **107**A and the male connector **136** of the adjoining frame piece **107**B.
(37) FIG. **5**A is a bottom view of the light fixture **101**, with an encircled detail area **500** to show context for the views of FIG. **5**B-D of the adjoining frame pieces **107**A-B, circuit board **113**, and the connector **135**.
(38) FIG. **5**B is a cutaway view of the encircled detail area **500** of FIG. **5**A and shows details of the adjoining frame pieces **107**A-B, circuit board **113**, light sources **115**A-N, and the connector **135**. As shown, the circuit board **113** includes one or more extra light sources **115**A-B clustered proximate the opening **141** of the connector **135**. Although not shown, the adjoining frame piece **107**A also includes electrical connections in a corner **145** to connect to the driver circuit **161**.
(39) Adding extra light sources **115**A-B can eliminate dark shadows at a corner **145** (e.g., connection point or connecting edge) between adjoining frame pieces **107**A-B, for example, at the bar connection point. As shown, doubling up or otherwise increasing a population density of light sources **115**A-N near the corner **145** of where the adjoining frame pieces **107**A-B abut each other helps eliminate dark shadows in the corners and provides an appearance of continuous illumination. The combination of the extra light sources **115**A-B and the connector **135** with the substantially transparent portion **139** and the opening **141** achieves the best aesthetics for the light fixture **101**. This combination provides the appearance of a light fixture **101** with continuous illumination by eliminating dark shadows and bright spots in the corner **145** of adjoining frame pieces **107**A-B.
(40) However, somewhat diminished aesthetics of the light fixture **101** may still be acceptable in some applications. Having just extra light sources **115**A-B without the substantially transparent portion **139** of the connector **135** may result in bright spots at the corner **145**, but can still reduce dark shadows. Also, having just an air gap instead of a substantially transparent portion **139** on the connector **135** may result in bright spots at the corner **145**, which is a different viewing artifact, but can still reduce dark shadows. Finally, if the connector **135** is formed of an entirely substantially opaque material, but with an opening **141**, this may result in a somewhat less dark shadow at the corner **145** of adjoining frame pieces **107**A-B.
(41) FIG. **5**C is a zoomed in view of the connector **135** that includes a substantially transparent portion **139** and a substantially opaque portion **137**. The substantially transparent portion **139** is an

optical part of the connector **135** to guide light and the substantially opaque portion **137** is a structural part of the connector **135** to connect the adjoining frame pieces **107**A-B and support the substantially transparent portion **139**.

(42) To form the connector **135**, the substantially transparent portion **139** can be formed integrally with the substantially opaque portion **137** (e.g., as one component or piece) by placing a substantially transparent material and a substantially opaque material in a mold. In this first example, the connector **135** can be injection molded as a single piece. Alternatively, the substantially transparent portion **139** and the substantially opaque portion **137** can be formed separately and then connected together. In this second example, the substantially transparent portion **139** can be configured to snap into the substantially opaque portion **137**. The connector **135** can be a mixture of the substantially opaque material and the substantially transparent material with the opening **141** to guide light through the connector **135** to eliminate dark shadows at the corner **145** (e.g., connection point) and have seamless and uniform illumination **117**.

(43) The size of the opening **141** of connector **135** may be relatively small and shaped like a slot due to manufacturing and application constraints. In order for the connector **135** to remain mechanically sound and yet provide as much light output as possible between the connection points of adjoining frame pieces **107**A-B, the size and shape may be constrained. Hence, there may be some limitations on how large the opening **141** can be in order to not sacrifice the mechanical structure of the connector **135**.

(44) FIG. **5**D is an isometric view of the assembled light fixture **101** with the connector **135** connecting the adjoining frame pieces **107**A-B together. As further shown, the L-shaped diffuser **111** is attached and covering the light sources **115**A-N of the circuit board **113**. Adding the extra light sources **115**A-B to increase the population density of light sources **115**A-N near the corner **145** (e.g., intersection) of adjoining frame pieces **107**A-B further reduces dark shadows.

(45) FIG. **6** is a cross-sectional view of the adjoining frame piece **107**A depicting details of the circuit board **113**, illumination light source **115**A, and a chamfer **151**. Each of the frame pieces **107**A-N of the light fixture **101** can include a chamfer **151** with an angled surface **153** configured to reflect illumination **117**. As shown, the chamfer **151** is on both sides of the adjoining frame piece **107**A and the bottom wall **129** of the adjoining frame piece **107**A is transparent where the L-shaped diffuser **111** is placed. The chamfer **151** is formed as a single piece on each frame piece **107**A-N. Chamfer **151** is structural and the extrusion holds the L-shaped diffuser **111** in place and the circuit board **113** down after the circuit board **113** is slid into the adjoining frame piece **107**A.

(46) Angled surface **153** of the chamfer **151** can be a beveled surface and can enable the light fixture **101** to achieve high optical efficiency. If the angled surface **153** is 90 degrees, then the light rays emitted from the light sources **115**A-N would bounce around and scatter in an interior cavity **157** and the optical efficiency may not be very high. Hence, the angled surface **153** can be less than 90 degrees to cause reflections of light rays emitted from the light sources **115**A-N to bounce off the angled surface **153** and reflect in a direction towards the bottom wall **129**. The light rays reflected off the angled surface **153** then pass through the bottom wall **129** of the L-shaped diffuser **111** as opposed to scattering around an interior cavity **157**.

(47) When light from the light sources **115**A-N strikes the chamfer **151**, reflectivity is increased which increases optical efficiency because more illumination **117** is outputted from the substantially transparent portions of the L-shaped diffuser **111**. In one example, a chamfer **151** formed of bare metal (e.g., aluminum) can achieve a light fixture **101** with an energy efficiency of 118 lumens per watt. This is a significant improvement over the normal energy efficiency requirement of 110 lumens per watt from the U.S. Department of Transportation (DOT) lighting standard.

(48) To further enhance optical efficiency, the angled surface **153** can include a highly reflective surface **155** formed at least in part by a paint, a coating, a laminate, or an anodized metal. Anodizing a base material, such as a metal (e.g., aluminum), that forms the chamfer **151** can create

a highly reflective surface **155** with an anodic oxide finish. Or the base material that forms the chamfer **151** can be painted with a highly reflective paint or include an added film to create the highly reflective surface **155**. The highly reflective surface **155** may be 90% or more reflective to light, 95% or more reflective to light, or 96% or more reflective to light to maximize light output (e.g., lumen performance) and thus optical efficiency. When the highly reflective surface **155** is 95% or more reflective to light, an energy efficiency of 125 lumens per watt can be achieved which exceeds the premium requirements of the DOT lighting standard.

(49) The angled surface **153** faces an interior cavity **157** between the light sources **115**A-N and the inside wall **127** of the L-shaped diffuser **111**. The angled surface **153** is configured to cause illumination **117** inside the interior cavity **157** to be emitted through substantially transparent portions of the inside wall **127** and the bottom wall **129** of the L-shaped diffuser **111**. The angled surface **153** is oriented at an angle of less than 90 degrees relative to the circuit board **113**. The chamfer **151** is co-extruded with the frame pieces **107**A-N and is configured to hold the L-shaped diffuser **111** in place and the circuit board **113** down.

(50) The terms “light fixture” or “luminaire” as used herein, are intended to encompass essentially any type of device that processes energy to generate or supply artificial light, for example, for general illumination of a space intended for use of occupancy or observation, typically by a living organism that can take advantage of or be affected in some desired manner by the light emitted from the device. However, a light fixture or luminaire may provide light for use by automated equipment, such as sensors/monitors, robots, etc. that may occupy or observe the illuminated space, instead of or in addition to light provided for an organism. However, it is also possible that one or more light fixtures or luminaires in or on a particular premises have other lighting purposes, such as signage for an entrance or to indicate an exit. In most examples, the light fixture(s) or luminaire(s) illuminate a space or area of a premises to a level useful for a human in or passing through the space, e.g., of sufficient intensity for general illumination of a room or corridor in a building or of an outdoor space such as a street, sidewalk, parking lot or performance venue. The actual source of illumination light in or supplying the light for a light fixture or luminaire may be any type of artificial light emitting device, several examples of which are included in the discussions below. Although the discussion herein is focused on light fixture types of luminaires that have a fixed position in a space, it should be understood that other types of luminaires can be used in lieu of light fixtures, such as lamps, in certain examples.

(51) The term “lighting system” as used herein, is intended to encompass essentially any type of system that either includes a number of such luminaires, such as light fixtures, coupled together for data communication and/or light fixture(s) coupled together for data communication with one or more control devices, such as wall switches, control panels, remote controls, central lighting or building control systems, servers.

(52) The illumination light output of a luminaire, such as a light fixture, for example, may have an intensity and/or other characteristic(s) that satisfy an industry acceptable performance standard for a general lighting application. The performance standard may vary for different uses or applications of the illuminated space, for example, as between residential, office, manufacturing, warehouse, or retail spaces. Any light fixture, however, may be controlled in response to commands received with the network technology of the lighting system, e.g. to turn ON/OFF, to dim the light intensity of the output, to adjust or tune color of the light output (for a light fixture having a variable color source), etc.

(53) Terms such as “illumination,” “artificial lighting,” or “illumination lighting” as used herein, are intended to encompass essentially any type of lighting that a device produces light by processing of electrical power to generate the light. A luminaire for an artificial lighting or illumination lighting application, for example, may take the form of a light fixture as described herein, a lamp, or other luminaire arrangement that incorporates a suitable light source, where the lighting device component or source(s) by itself contains no intelligence or communication

capability. The illumination light output of an artificial illumination type luminaire, such as a light fixture, for example, may have an intensity and/or other characteristic(s) that satisfy an industry acceptable performance standard for a general lighting application.

(54) Terms such as “lighting device” or “lighting apparatus,” as used herein, are intended to encompass essentially any combination of an example of a luminaire, such as the light fixture, discussed herein with other elements such as electronics and/or support structure, to operate and/or install the particular luminaire implementation. Such electronics hardware, for example, may include some or all of the appropriate driver(s) for the illumination light sources, any associated control processor or alternative higher level control circuitry, and/or data communication interface(s). The electronics for driving and/or controlling the light sources may be incorporated within the luminaire or located separately and coupled by appropriate means to the light source component(s).

(55) The term “coupled” as used herein refers to any logical, optical, physical, or electrical connection, link or the like by which signals or light produced or supplied by one system element are imparted to another coupled element. Unless described otherwise, coupled elements or devices are not necessarily directly connected to one another and may be separated by intermediate components, elements, or communication media that may modify, manipulate, or carry the light or signals

(56) The orientations of the lighting system **100**, lighting device **160**, associated components, and/or any complete devices incorporating the light fixture **101**, such as shown in any of the drawings, are given by way of example only, for illustration and discussion purposes. In operation, the light fixture **101** may be oriented in any other direction suitable to the particular application, for example upright, sideways, or any other orientation. Also, to the extent used herein, any directional term, such as lateral, longitudinal, up, down, upper, lower, top, bottom, above, below, front, rear, side, left, and right are used by way of example only, and are not limiting as to direction or orientation of any light fixture **101** or component of the light fixture **101** constructed as otherwise described herein.

(57) Unless otherwise stated, any and all measurements, values, ratings, positions, magnitudes, sizes, angles, and other specifications that are set forth in this specification, including in the claims that follow, are approximate, not exact. Such amounts are intended to have a reasonable range that is consistent with the functions to which they relate and with what is customary in the art to which they pertain. For example, unless expressly stated otherwise, a parameter value or the like may vary by as much as ±5% or as much as ±10% from the stated amount. The terms “approximately,” “significantly,” or “substantially” mean that the parameter value or the like varies up to +25% from the stated amount. The term “substantially transparent” means 85% or more transmissive to light. The term “substantially opaque” means 20% or less transmissive to light. The term “partially substantially transparent” means a portion is substantially transparent and another portion is substantially opaque. The term “highly reflective” means 90% or more reflective to light.

(58) It will be understood that the terms and expressions used herein have the ordinary meaning as is accorded to such terms and expressions with respect to their corresponding respective areas of inquiry and study except where specific meanings have otherwise been set forth herein. Relational terms such as first and second and the like may be used solely to distinguish one entity or action from another without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “containing,” “contain”, “contains,” “with,” “formed of,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises or includes a list of elements or steps does not include only those elements or steps but may include other elements or steps not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “a” or “an” does not, without further constraints, preclude the existence of additional identical elements in the process, method,

article, or apparatus that comprises the element.
(59) In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various examples for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed examples require more features than are expressly recited in each claim. Rather, as the following claims reflect, the subject matter to be protected lies in less than all features of any single disclosed example. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.
(60) While the foregoing has described what are considered to be the best mode and/or other examples, it is understood that various modifications may be made therein and that the subject matter disclosed herein may be implemented in various forms and examples, and that they may be applied in numerous applications, only some of which have been described herein. It is intended by the following claims to claim any and all modifications and variations that fall within the true scope of the present concepts.
(61) The scope of protection is limited solely by the claims that now follow. That scope is intended and should be interpreted to be as broad as is consistent with the ordinary meaning of the language that is used in the claims when interpreted in light of this specification and the prosecution history that follows and to encompass all structural and functional equivalents. Notwithstanding, none of the claims are intended to embrace subject matter that fails to satisfy the requirement of Sections 101, 102, or 103 of the Patent Act, nor should they be interpreted in such a way. Any unintended embracement of such subject matter is hereby disclaimed.

Claims

1. A light fixture, comprising: a frame that includes a plurality of frame pieces configured for assembly around a ceiling tile to form the light fixture, each of the plurality of frame pieces including an L-shaped diffuser and a circuit board that includes a plurality of light sources to emit illumination; wherein: the L-shaped diffuser includes a diffuser body that is asymmetric in transparency to direct more illumination from the light sources to inside the ceiling tile than outside the ceiling tile; the diffuser body includes an outside wall that is substantially opaque, an inside wall that is at least partially substantially transparent, and a bottom wall that is substantially transparent; and the inside wall includes an upper portion that is substantially opaque and a lower portion that is substantially transparent.
2. The light fixture of claim 1, wherein: the lower portion is configured to illuminate inside the ceiling tile and the upper portion is configured to direct illumination to the bottom wall.
3. The light fixture of claim 1, wherein: the outside wall is substantially opaque to block illumination from appearing outside of the ceiling tile and thereby reduce glare.
4. The light fixture of claim 1, wherein: the diffuser body is co-extruded as a single extrusion of optical material with a white pigment for substantially opaque portions.
5. The light fixture of claim 4, wherein: the optical material is formed of polycarbonate or acrylic.
6. The light fixture of claim 1, further comprising: a connector to connect adjoining frame pieces of the plurality of frame pieces together; wherein the connector includes a substantially opaque portion below the circuit board and a substantially transparent portion above the circuit board to allow illumination to pass between the adjoining frame pieces.
7. The light fixture of claim 6, wherein the substantially transparent portion includes an opening to create a light guide between the adjoining frame pieces.
8. The light fixture of claim 7, wherein the light guide of the substantially transparent portion is configured to reduce a shadow in a corner where the adjoining frame pieces are connected together.
9. The light fixture of claim 7, wherein the circuit board includes one or more extra light sources clustered proximate the opening.

10. The light fixture of claim 6, wherein the connector is injection molded as a single piece.
11. The light fixture of claim 6, wherein the substantially transparent portion is configured to snap into the substantially opaque portion.
12. The light fixture of claim 1, where each of the frame pieces includes a chamfer with an angled surface configured to reflect illumination.
13. The light fixture of claim 12, where the angled surface includes a highly reflective surface formed at least in part by a paint, a coating, a laminate, or an anodized metal.
14. The light fixture of claim 12, where the angled surface faces an interior cavity between the light sources and the inside wall of the L-shaped diffuser.
15. The light fixture of claim 14, where the angled surface is configured to cause illumination inside the interior cavity to be emitted through substantially transparent portions of the inside wall and the bottom wall of the L-shaped diffuser.
16. The light fixture of claim 12, where the angled surface is oriented at an angle of less than 90 degrees relative to the circuit board.
17. The light fixture of claim 12, where the chamfer is co-extruded with the frame pieces and is configured to hold the L-shaped diffuser in place and the circuit board down.
18. A lighting device, comprising: a light fixture including a frame that includes a plurality of frame pieces configured for assembly around a ceiling tile to form the light fixture, each of the plurality of frame pieces including an L-shaped diffuser and a circuit board that includes a plurality of light sources to emit illumination, wherein the L-shaped diffuser includes a diffuser body that is asymmetric in transparency to direct more illumination from the light sources to inside the ceiling tile than outside the ceiling tile; and a connector to connect adjoining frame pieces of the plurality of frame pieces together, wherein the connector includes a substantially opaque portion below the circuit board and a substantially transparent portion to create a light guide above the circuit board to allow illumination to pass between the adjoining frame pieces.
19. The lighting device of claim 18, where each of the frame pieces includes a chamfer with an angled surface configured to reflect illumination.
20. The lighting device of claim 18, further comprising: a driver circuit to control the light sources; and a cable configured to connect via electrical connection to the circuit board and the driver circuit to control the light sources; wherein the driver circuit is configured to mount on a ceiling tile grid and is mounted separate from the light fixture assembled around the ceiling tile.
21. A light fixture, comprising: a frame that includes a plurality of frame pieces configured for assembly around a ceiling tile to form the light fixture, each of the plurality of frame pieces including an L-shaped diffuser and a circuit board that includes a plurality of light sources to emit illumination; wherein: the L-shaped diffuser includes a diffuser body that is asymmetric in transparency to direct more illumination from the light sources to inside the ceiling tile than outside the ceiling tile; the diffuser body includes an outside wall that is substantially opaque, an inside wall that is at least partially substantially transparent, and a bottom wall that is substantially transparent; the outside wall and the inside wall are parallel and vertical; and the bottom wall is perpendicular to the outside wall and the inside wall.